

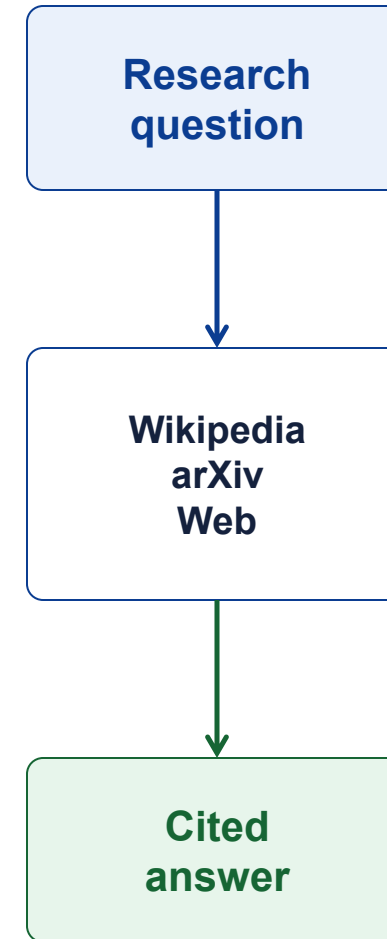
Async Research Assistant

A resilient, concurrent system for cited research answers

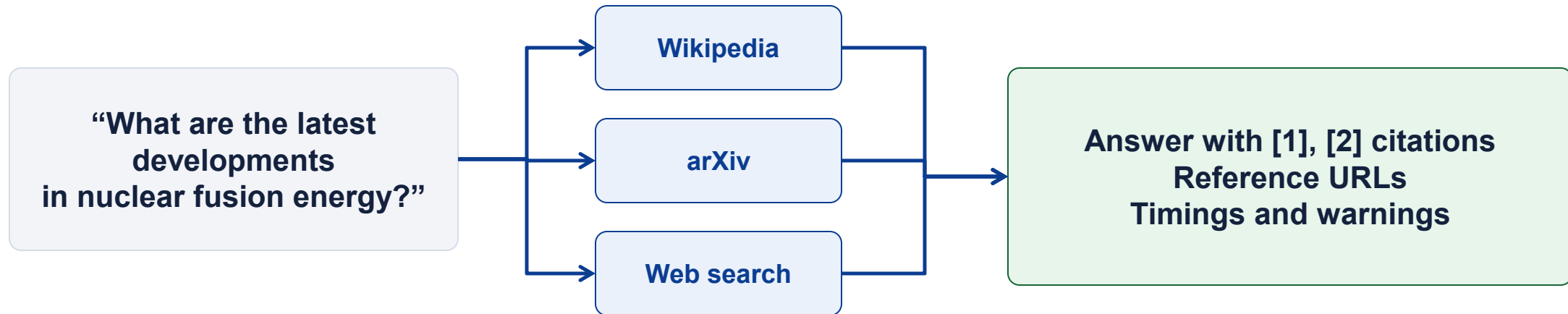
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M301 Software Engineering Final Project

github.com/Hasanov57/AI-Academy---SE---Final-Project---AI-Researcher



One question becomes one concise answer grounded in retrieved evidence

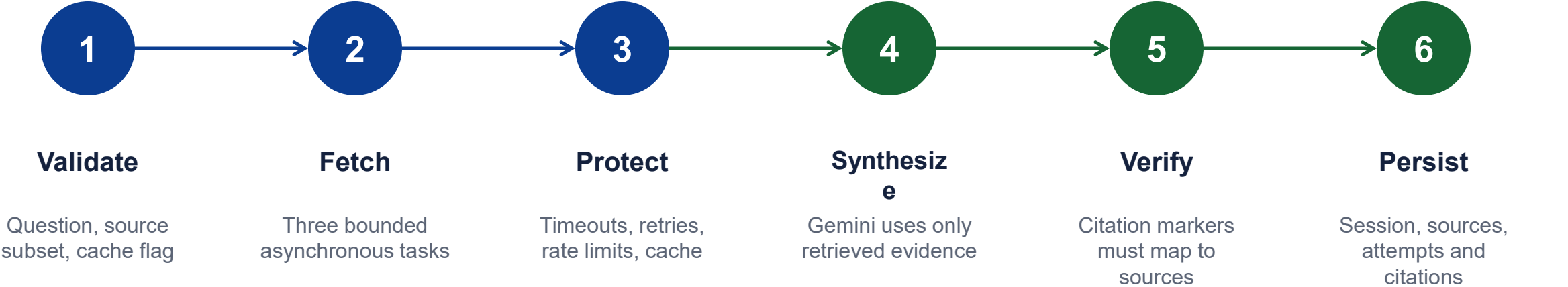


Two entry points

CLI for grading and automation

Streamlit UI for project-day interaction

Users can select sources and bypass cached results when freshness matters.

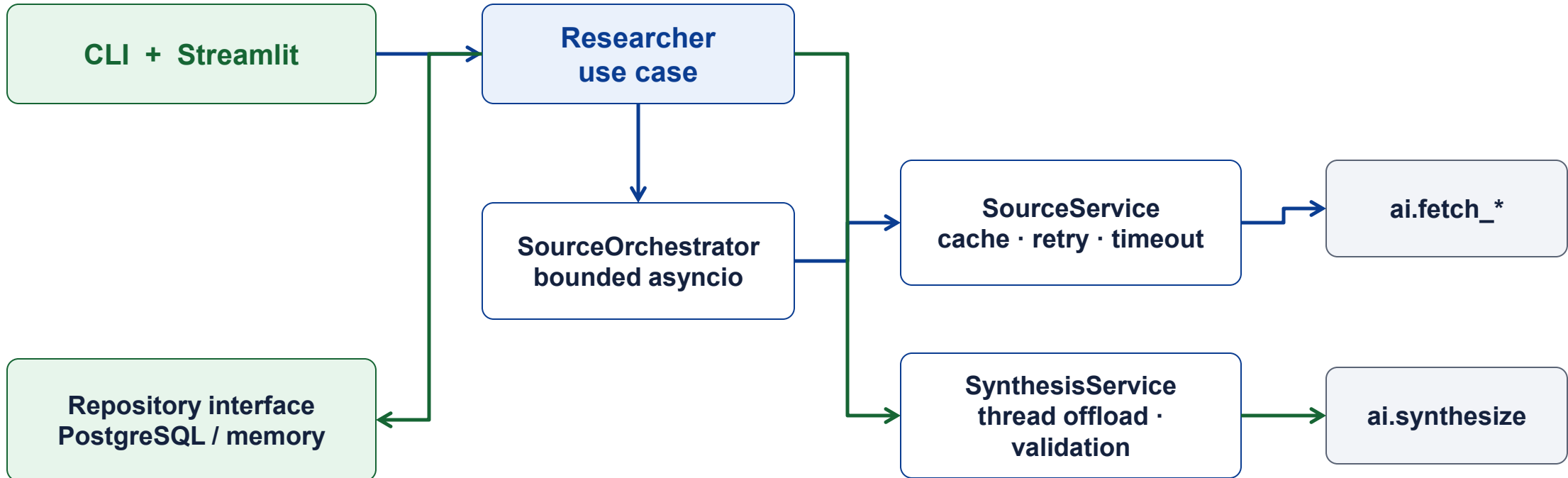


Graceful degradation

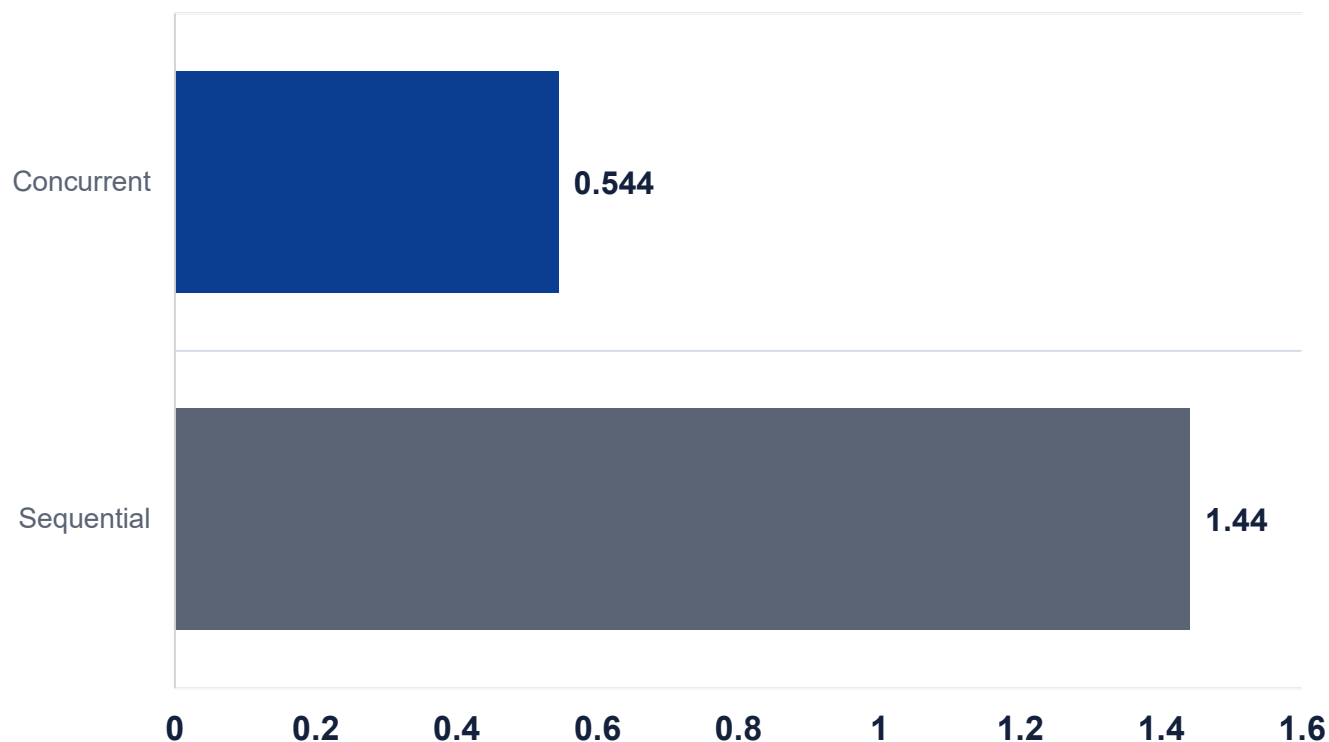
A failed source becomes a warning. The remaining evidence still reaches synthesis; all-source failure stops cleanly before the LLM call.

Student-owned software engineering layer

Provided AI package



Composition keeps provider code replaceable in tests. One repository contract supports durable PostgreSQL and fast in-memory execution.

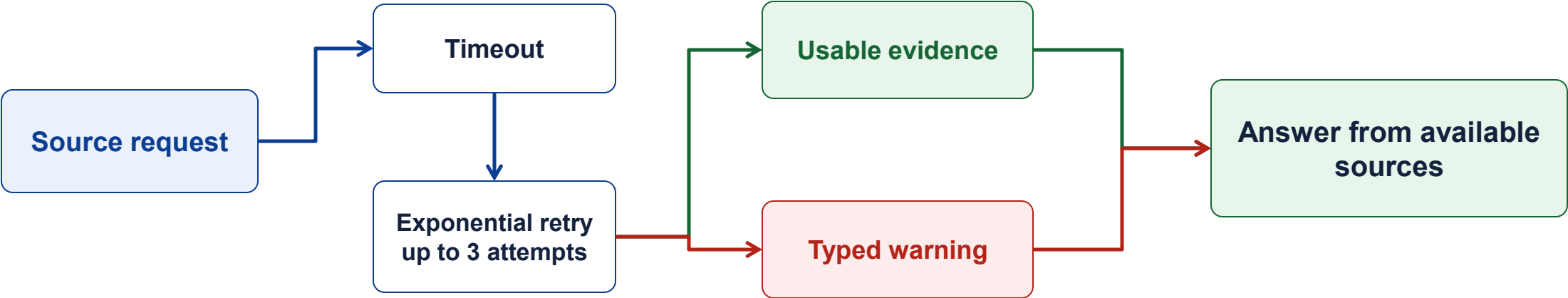


2.64×
median speedup

- Three live sources
- Same question and machine
- Application cache bypassed
- Retrieval measured separately from synthesis

Why asyncio? Source calls spend most of their time waiting for remote servers. Overlapping that wait reduces wall-clock time without multiprocessing overhead.

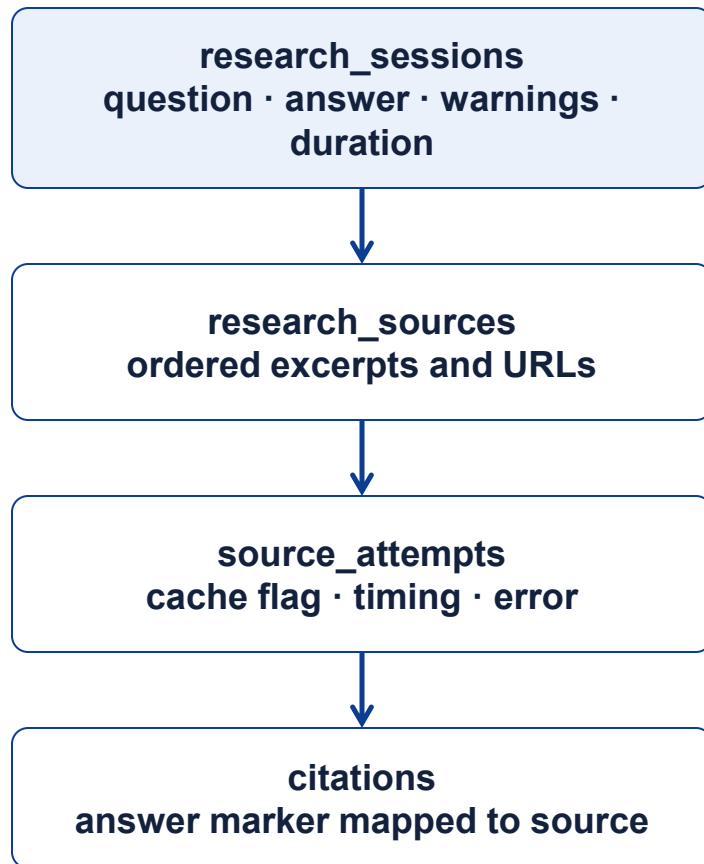
External calls are unreliable; the pipeline makes failures visible and contained



Citation safety

The validator rejects missing, dangling, or mismatched numeric references and retries synthesis with the specific validation error.

PostgreSQL keeps an audit trail for every research session



TTL source cache



- Fast repeated questions
- Expired rows removed lazily
- Bypass cache for fresh demonstrations
- Memory repository for offline tests

Docker Compose starts the application and PostgreSQL with a database health check.

Deploy ⋮

Research settings

- ☒ Wikipedia
- ☒ arXiv
- ☒ Web search
- ☐ Bypass cache

Async Research Assistant

Ask one research question. The assistant searches selected sources in parallel and returns a concise answer with citations.

Research question

What are the latest developments in nuclear fusion energy?

Research

Demo plan

- 1 Ask one current research question
- 2 Show three sources starting together
- 3 Match inline markers to reference links
- 4 Point out timing and session ID

Backup: the scripted five-question demo under scripts/demo.py

64

offline tests passing

73.56%

branch-aware coverage

0

live-network test
dependencies

PASS

Ruff and strict mypy

Current limitations

- Citation structure is checked, but semantic support is not proven
- Rate limits coordinate within one process, not across replicas
- Source relevance depends on upstream search quality

Next improvements

- Semantic citation entailment checks
- Multi-provider failover
- Distributed rate-limit coordination

Samir Həsənov

Core workflow and concurrency
CLI, rendering, and Streamlit UI

Elmir Əsgərov

Source and LLM service wrappers
Retries, rate limits, and synthesis

Ramil Məmmədəliyev

Storage and configuration
Scripts, packaging, and Docker

Questions?

We can explain the architecture, trade-offs, and request flow, then demonstrate the system live.

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